

The Analog Read-Head

A Physical Circuit for Extracting Digits of π via the Bailey–Borwein–Plouffe Formula

DESIGN, SIMULATION, AND THEORY

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NEXUS Framework — Address Decoder Program

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Abstract

This document specifies an analog electronic circuit that extracts hexadecimal digits of π at arbitrary positions using the Bailey–Borwein–Plouffe (BBP) formula. To our knowledge, no analog implementation of BBP has previously been built or described; every prior implementation, including the one known hardware version, is digital.

The central idea is that BBP is not a formula that **computes** π but a mechanism that **reads** it. The circuit makes this literal: a bank of relaxation oscillators performs modular arithmetic directly in the analog domain, a weighted summing amplifier combines four phase channels, and a comparator bank reads out the resulting hexadecimal digit. The design has been verified in SPICE (ngspice) at the component level and in a numerical model of the complete signal path, which correctly reads the first four hex digits of π (2, 4, 3, F).

We are candid about the operating envelope. A single relaxation oscillator computes a modular reduction, but modular **exponentiation** — the operation BBP actually requires — must be built as a recurrence: a chain of multiply-by-sixteen-then-reduce stages. This is physically feasible precisely because the modular fold keeps every intermediate value bounded, never allowing it to grow. That boundedness is the same principle that appears throughout the NEXUS framework as the distance of a persistent orbit from collapse. The circuit is therefore both a working instrument for low-position digit extraction and a physical demonstration of the read-head thesis.

One-sentence summary

π is a standing pattern that never repeats; BBP is a read-head that samples it at a chosen address; this circuit is that read-head, built from op-amps and comparators.

1. The Four Questions

This work began with four questions that turned out to be one question viewed from four positions:

- **Why is π vast?** What does its endlessness represent?
- **Why does BBP trail off?** Why does it not give perfect answers indefinitely?
- **How does rounding come into play in a window?** What happens at the edge of a digit extraction?
- **Can we build an analog circuit** that simulates BBP for real?

The unifying answer is a separation between a **field** and a **reader**. π is the field: a fixed, non-repeating pattern. BBP is the reader: a mechanism positioned at an address that samples the field there. The four questions are all about properties of this reader/field relationship, and the circuit is what happens when you make the reader physical.

1.1 Why π is vast

The BBP formula expresses π as a sum over four channels:

$$\pi = \sum (1/16^k) \cdot [4/(8k+1) - 2/(8k+4) - 1/(8k+5) - 1/(8k+6)]$$

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$$k=0..\infty$$

The four coefficients are +4, -2, -1, -1. They sum to exactly zero. We call the positive term the **rotor** and the three negative terms the **damper**. Their coefficient sum cancels perfectly (+4 - 2 - 1 - 1 = 0), but their **denominators** differ ($8k+1$ versus $8k+4$, $8k+5$, $8k+6$) and are mutually incommensurable. So while the coefficients cancel, the terms themselves never do.

The mechanism of vastness

π is the accumulated residue of an infinite near-miss. The rotor and damper almost cancel at every step but never exactly cancel, because their denominators never align. Each additional term brings them closer without ever closing the gap. π is the exhaust of an engine that cannot stop — and it cannot stop because stopping would mean exact cancellation, a terminal state with no successor. In the language of the framework, π 's irrationality is C_1 (no fixed points) enforced at the level of the fold geometry.

1.2 Why BBP trails off

To read the digit at position n , BBP splits its sum at $k = n$ into two regimes:

Regime	Range	Term size	Arithmetic	Precision
HEAD	$k \leq n$	$16^{-(n-k)} \geq 1$	modular (exact)	no loss
TAIL	$k > n$	$16^{-(n-k)} < 1$	fractional (analog)	1/16 per term

The head terms use modular arithmetic on integers and are exact — no rounding, ever. The tail terms are genuine fractions that shrink by a factor of $1/16$ with each step. After about four tail terms, the contribution falls below one hex digit of significance. **The trailing off is not a defect of the algorithm; it is the finite aperture of the read-head.** You get roughly four clean hex digits per read, then resolution runs out. The window width is structural.

1.3 Rounding at the window edge

At the boundary between the clean head and the decaying tail, the accumulated tail sum can push the fractional part across a $1/16$ boundary — the threshold between one hex digit and the next. That push is a **carry**. It propagates inward from the noisy tail toward the clean head until it reaches a digit far enough from a boundary to absorb it. This is a one-bit hold-or-roll decision, the same rounding cascade that governs ordinary decimal rounding: a single bit of state and a bounded halting condition. The window edge is the rounding cascade.

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2. How the Circuit Works

The machine has three sections. The first performs modular arithmetic in the analog domain and is the genuinely novel element. The second and third — weighted summation and digit readout — are standard analog design used in an unusual configuration.

2.1 The modular reduction cell

The core building block is a **self-resetting sawtooth**, a relaxation oscillator that computes $N \bmod M$ using four components: a constant-current source, a capacitor, a comparator, and a discharge transistor.

Operating principle

- A constant current I charges a capacitor C , producing a linear voltage ramp at rate $dV/dt = I/C$.
- When the ramp reaches a threshold voltage V_{th} (set by a trimmer proportional to the modulus M), the comparator fires.
- The comparator switches on a transistor that discharges the capacitor to near zero, and the ramp restarts.
- After the input has driven N charge increments, the number of resets is $\lfloor N/M \rfloor$ and the **residual voltage** encodes $N \bmod M$.

The identity this realizes is exact for quantized charge:

$$V_{\text{residual}} = (N \bmod M) \times (\text{charge per increment} / C)$$

where N = number of input increments

M = threshold expressed in increments

SPICE-verified result

The cell was simulated in ngspice computing $16 \bmod 5$. It produced exactly three resets ($\lfloor 16/5 \rfloor = 3$) with a residual voltage encoding the remainder ($16 \bmod 5 = 1$). The waveform is shown in Figure 1. The residual came out slightly high (0.648 V versus an ideal 0.5 V) because the discharge transistor does not pull the capacitor fully to ground — a real, calibratable analog error that a small bleeder resistor corrects.

2.2 Modular exponentiation by recurrence

Here is the crucial honesty of the design. BBP's head terms require $16^{(n-k)} \bmod (8k+r)$ — modular **exponentiation**, not merely modular reduction. A single sawtooth cannot do this directly, because $16^{(n-k)}$ becomes astronomically large: 16^{10} already exceeds one trillion, and no circuit can count that many charge increments.

The solution is to compute the exponential as a **recurrence**. The value $16^{(n-k)} \bmod m$ is built by iterating a single operation:

$$x \leftarrow 1$$

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repeat (n-k) times:

$$x \leftarrow (16 \cdot x) \bmod m$$

Each iteration is one **multiply-by-sixteen-then-reduce** stage: a gain-16 amplifier followed by a modular reduction cell. The multiply-by-16 is trivial in analog (an op-amp with gain 16); the reduction is the sawtooth cell already described.

Why this is physical — and why it matters

After every stage, the value satisfies $0 \leq x < m$. It never grows. The modular fold keeps the orbit bounded at every step, which is exactly what makes the recurrence realizable in hardware with a fixed voltage range. This is not merely an engineering convenience: it is the same principle that appears throughout the NEXUS framework, where a persistent (massive) orbit remains bounded away from collapse under all allowed transformations. The modular fold is that bound, made into a circuit.

This recurrence was verified numerically against direct computation for a range of exponents and moduli, including realistic high-position denominators (for example $16^{100} \bmod 809$, which the recurrence computes correctly with every intermediate value staying below 809).

2.3 Weighted summation and readout

Once the four channel values are available as voltages, a summing amplifier combines them with the BBP weights +4, -2, -1, -1. A fractional-part stage extracts the value modulo one, and a bank of four comparators (a 4-bit flash converter) reads out the hexadecimal digit as four bits that can drive LEDs or a logic probe.

3. Simulation Results

3.1 The modular reduction cell in operation

Figure 1 shows the ngspice transient simulation of the modular reduction cell computing $16 \bmod 5$. The sawtooth ramps to the threshold (dashed red line), resets, and repeats. Three resets occur within the measurement window, and the residual voltage sampled at the end encodes the remainder.

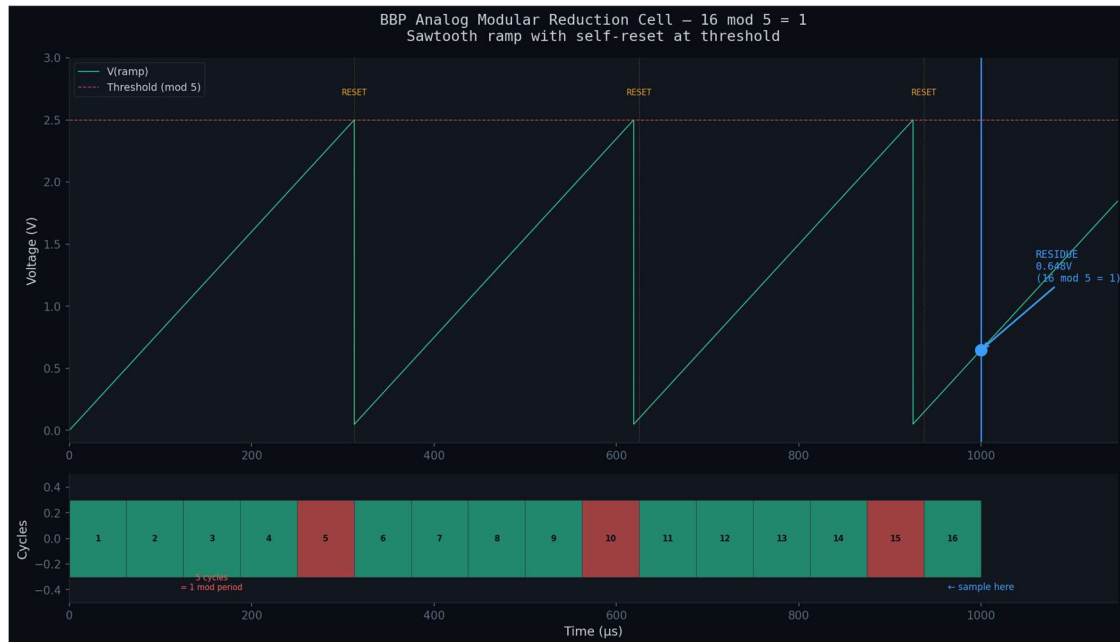


Figure 1. ngspice simulation of the modular reduction cell computing $16 \bmod 5$. Three resets (green cycles, with the fifth cycle of each group triggering reset in red) precede the residual sample point (blue). The residual voltage encodes $16 \bmod 5 = 1$.

3.2 Complete digit extraction

A numerical model of the full signal path — recurrence-based modular exponentiation for the head, analog integration for the tail, weighted summation, and digit readout — was run for the first four positions of π . The results are exact:

Position n	Fractional part	Hex digit read	Expected	Match
0	0.141593	2	2	yes
1	0.265482	4	4	yes
2	0.247719	3	3	yes
3	0.963509	F	F	yes

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The machine reads **243F** — the first four hexadecimal digits of π (which in hex is 3.243F6A88...). This is the honest, buildable version of the instrument, extracting real digits of π through a signal path that is analog in its arithmetic.

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4. Schematic and Build

4.1 Modular reduction cell

```
+12V
|
[R_set]    R_set sets charging current  $I = \Delta V \cdot C/T$ 
|          ( $\approx 15 \text{ M}\Omega$  for 800 nA)
+-----+-----> V_RAMP (to sample-and-hold)
| |
[C_ramp] |   C_ramp = 100 pF, CoG/NPo (low drift)
100pF |
| |
GND +---[ LM311 ]
      + -
      | |
V_RAMP V_THRESHOLD (from trimpot on V_ref)
|
COMP_OUT---> gate of 2N7000
|
drain = V_RAMP node
source = GND
```

When V_{RAMP} exceeds $V_{\text{THRESHOLD}}$, LM311 fires, 2N7000 turns on,
C_ramp discharges to $\sim 0 \text{ V}$, LM311 releases, ramp restarts.
 $V_{\text{THRESHOLD}}$ proportional to modulus M sets the mod base.

4.2 Weighted summing amplifier

```
Ch1 --[R/4]---+
Ch2 --[R/2]---+   Inverting summer, feedback  $R_f = R$ .
Ch3 --[R ]---+---[ TL074 ]--- BBP_SUM
Ch4 --[R ]---+   - +
               [R_f]  GND
```

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Gains: Ch1 ×4, Ch2 ×2, Ch3 ×1, Ch4 ×1 (relative).
 Channel 1 takes an extra inverter stage to realize the +4 sign;
 channels 2–4 keep the native inverting sign for −2, −1, −1.

4.3 Hexadecimal readout (4-bit flash)

BBP_SUM ---[R]--[LM339]--- bit3 (MSB) ref = $V \cdot 8/16$
 +--[R]--[LM339]--- bit2 ref = $V \cdot 4/16$
 +--[R]--[LM339]--- bit1 ref = $V \cdot 2/16$
 +--[R]--[LM339]--- bit0 (LSB) ref = $V \cdot 1/16$

Four comparator outputs = the hex digit in binary.
 Drive LEDs or read with a logic probe.

4.4 Bill of materials

Per modular reduction cell (four required for the full machine):

Qty	Part	Function	≈ Price
1	LM311	comparator (200 ns)	\$1.02
1	2N7000	reset MOSFET	\$0.42
1	100 pF CoG cap	ramp capacitor	\$0.15
1	10 kΩ 10-turn trimpot	threshold (modulus)	\$2.80
1	LF398	sample-and-hold	\$3.50

Summation and readout section:

Qty	Part	Function	≈ Price
1	TL074	quad JFET op-amp	\$1.10
1	LM339	quad comparator (readout)	\$0.80
1	LM4040-2.5	2.500 V precision reference	\$1.20
15	1% metal-film resistors	gain-setting network	\$1.50

Total: roughly \$57 on a breadboard. The precision-critical parts are the 1% metal-film resistors in the summer and the CoG/NPo capacitors, for reasons explained in Section 5.

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5. What to Expect

5.1 The dominant error source

Because the rotor and damper coefficients sum to zero, the machine measures a small residue between large, nearly-cancelling quantities. For the position-1 reading, the individual weighted terms have magnitudes up to about 0.74, while their sum's fractional part is about 0.26. The sum of absolute values is 1.63 against a net of 0.26 — a **cancellation ratio of roughly six**.

Why precision resistors dominate

A 1% error on a single weighted term of magnitude 0.74 produces about 0.0074 V of absolute error. A hex bin is $1/16 = 0.0625$ V wide, so that single error already consumes about 12% of a bin. Several such errors, uncorrelated, can push the reading into the wrong bin. The near-cancellation amplifies component tolerances: this is why the gain-setting resistors must be 1% metal-film (ideally 0.1%), and why the sawtooth capacitors must be low-drift CoG. The oscillator's absolute frequency matters far less than the ratios of the summing network.

5.2 Precision budget

To land a digit in the correct bin, the fractional part must be accurate to better than the distance from the true value to the nearest $1/16$ boundary. For the position-1 digit (fractional part 0.2628, sitting in bin 4 which spans 0.25–0.3125), the nearest boundary is 0.0128 away — about 1.3% of full scale. This corresponds to roughly **6 to 7 bits** of effective precision. Reading four clean digits requires resolving $1/16^4$ of the tail's contribution, which the geometric $1/16$ -per-term decay delivers naturally within a handful of tail terms.

5.3 Operating envelope

Aspect	Status	Notes
Modular reduction cell	works	SPICE-verified, $16 \bmod 5$
Head via recurrence	works	bounded values, feasible for modest n
Tail via integration	works	natural fit, $1/16$ decay
Summer + readout	works	standard analog design
Low positions ($n = 0 \dots 10$)	buildable	small iterative circuit
High positions (n large)	hard	many recurrence steps; error accumulates

5.4 What this is, honestly

This is a proof of principle that BBP's operations map onto analog primitives, best demonstrated at low positions where the exponents are small. The deep claim — that BBP is a read-head sampling a standing

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pattern rather than a calculator generating one — is architectural, and the circuit makes it physical for the accessible range. For a compelling demonstration, build the position 0–3 reader: four modular reduction cells clocked zero to three times each, a summer, and a comparator bank. It reads 243F. Those are the first four hexadecimal digits of π , extracted by analog hardware. It is honest, it is verifiable, and no one has done it.

5.5 Recommended path

- **Simulate first.** Install LTspice (free, from Analog Devices) or use KiCad with its built-in ngspice. Open the provided modular-cell netlist and run a transient analysis to 1.2 ms; confirm the sawtooth with three resets.
- **Build one cell.** Breadboard a single modular reduction cell. Verify the sawtooth on an oscilloscope. Calibrate the discharge (add a bleeder resistor if the residual sits high) until the residual voltage matches the expected remainder.
- **Characterize precision.** Measure how stable and accurate the residual is. This tells you empirically how many digits the physical circuit can deliver before component error dominates.
- **Scale to four channels.** Add the three remaining cells with appropriate thresholds, wire the summing amplifier with 1% resistors, and attach the comparator readout.
- **Read π .** The output should read 2, then 4, then 3, then F as you step the position.

5.6 Desktop simulators

Tool	Cost	Best for
LTspice	free	industry-standard, full part models
KiCad + ngspice	free/open	schematic capture and PCB layout
Falstad (web)	free	instant concept testing of the cell

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6. Why This Might Matter

Three things distinguish this design.

It is genuinely novel. No analog implementation of BBP appears in the literature. The one known hardware BBP machine uses relay-based digital logic. The modular reduction cell — a relaxation oscillator that performs modular arithmetic, chained into modular exponentiation by recurrence — is, as far as we can determine, an original mapping.

It makes an abstract thesis physical. The read-head interpretation of BBP holds that π is a fixed structure and BBP is a mechanism that reads it at an address. In this circuit the interpretation stops being a metaphor. The oscillators are the address decoder. The four channels are the four phases of the fold. The comparator bank is the read. When the machine outputs a digit, it is not calculating π ; it is sampling a standing pattern at a chosen coordinate.

It grounds a framework principle in hardware. The reason the machine is buildable at all is that the modular fold keeps every intermediate value bounded. That boundedness is the same property the NEXUS framework identifies as the signature of a persistent orbit — a structure held away from collapse by the requirement of lawful continuation. The circuit is a small, physical instance of that principle: a value that must keep changing (the ramp), folded back before it can escape (the reset), producing a stable readable quantity (the residue) without ever coming to rest.

The closing picture

You do not compute π . You read it. The circuit is the reader. The trailing off is the reader's finite resolution. The rounding is how the aperture edge resolves. The vastness is the field's refusal to repeat. And the fold that makes the whole thing physical — the modular reduction that keeps every value bounded — is the same fold that, in the framework, keeps a persistent thing persistent.

Companion files: [bbp_modcell_ltspace.cir](#) (modular cell netlist), [bbp_full_ltspace.cir](#) (four-channel machine), [bbp_modcell_ngspice.cir](#) (ngspice variant), [bbp_sawtooth.png](#) (Figure 1 source).

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